

Industrial Internship Report on IoT-Based Precision Irrigation Management System Using ESP32

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Executive Summary

This report presents the work completed during the Industrial Internship conducted by upskill Campus and The IoT Academy in collaboration with UniConverge Technologies Pvt. Ltd. (UCT). The internship provided practical exposure to Industrial Internet of Things (IIoT) concepts and embedded system development through a real-world project completed over six weeks.

The project undertaken was **"IoT-Based Precision Irrigation Management System Using ESP32."** The objective of the project was to develop an intelligent irrigation system capable of monitoring soil moisture, temperature, humidity, rainfall, and light intensity to automate irrigation decisions. The system uses an ESP32 microcontroller, DHT22 sensor, soil moisture sensor, rain sensor, LDR, OLED display, LED, and buzzer to monitor environmental conditions and operate the irrigation pump efficiently.

The project was designed and simulated using the Wokwi platform. The implemented system helps reduce water wastage by irrigating only when soil moisture is low, no rainfall is detected, sufficient daylight is available, humidity is below the defined threshold, and temperature is above the required level.

This internship provided valuable hands-on experience in IoT system design, embedded programming using Arduino IDE, sensor interfacing, decision-based automation, and project documentation. It also strengthened my problem-solving, debugging, and technical reporting skills, making it an enriching learning experience.

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1 Preface

The Industrial Internship conducted by upskill Campus, The IoT Academy, and UniConverge Technologies Pvt. Ltd. provided me with an excellent opportunity to gain practical exposure to Embedded Systems and the Internet of Things (IoT). Throughout this six-week internship, I developed both theoretical knowledge and practical skills that strengthened my understanding of modern embedded technologies and their real-world applications.

During the initial weeks of the internship, I built a strong foundation in IoT architecture, communication protocols, embedded hardware, microcontrollers, sensors, actuators, circuit design, and industrial IoT applications. The training also introduced me to networking concepts, cloud connectivity, embedded system design, and the engineering approach followed in developing reliable IoT solutions. These learning experiences prepared me to apply the acquired concepts in a practical project.

As part of the internship, I developed an IoT-Based Precision Irrigation Management System Using ESP32. The project focused on designing an intelligent irrigation system capable of monitoring soil moisture, temperature, humidity, rainfall, and light intensity to automate irrigation decisions. Through this project, I gained hands-on experience in ESP32 programming, sensor interfacing, embedded software development, system testing, and simulation using the Wokwi platform.

The internship followed a structured learning approach that began with understanding fundamental concepts before progressing to practical implementation. This methodology helped me understand not only how embedded systems and IoT technologies function individually, but also how they integrate to solve real-world engineering problems.

Throughout this internship, I improved my technical knowledge, problem-solving ability, debugging skills, and technical documentation. Working on both theoretical concepts and practical implementation increased my confidence in developing embedded IoT applications and strengthened my interest in this field.

I would like to express my sincere gratitude to **UniConverge Technologies Pvt. Ltd. (UCT), upskill Campus, The IoT Academy, my college faculty, and all the mentors** for their constant guidance, encouragement, and support throughout the internship. Their valuable inputs made this internship an enriching learning experience.

I encourage my juniors and peers to actively participate in internship opportunities, as they provide practical exposure, enhance technical skills, and bridge the gap between academic learning and real-world engineering practice.

2 Introduction

2.1 About UniConverge Technologies Pvt Ltd

A company established in 2013 and working in Digital Transformation domain and providing Industrial solutions with prime focus on sustainability and RoI.

For developing its products and solutions it is leveraging various **Cutting Edge Technologies** e.g. **Internet of Things (IoT), Cyber Security, Cloud computing (AWS, Azure), Machine Learning, Communication Technologies (4G/5G/LoRaWAN), Java Full Stack, Python, Front end** etc.



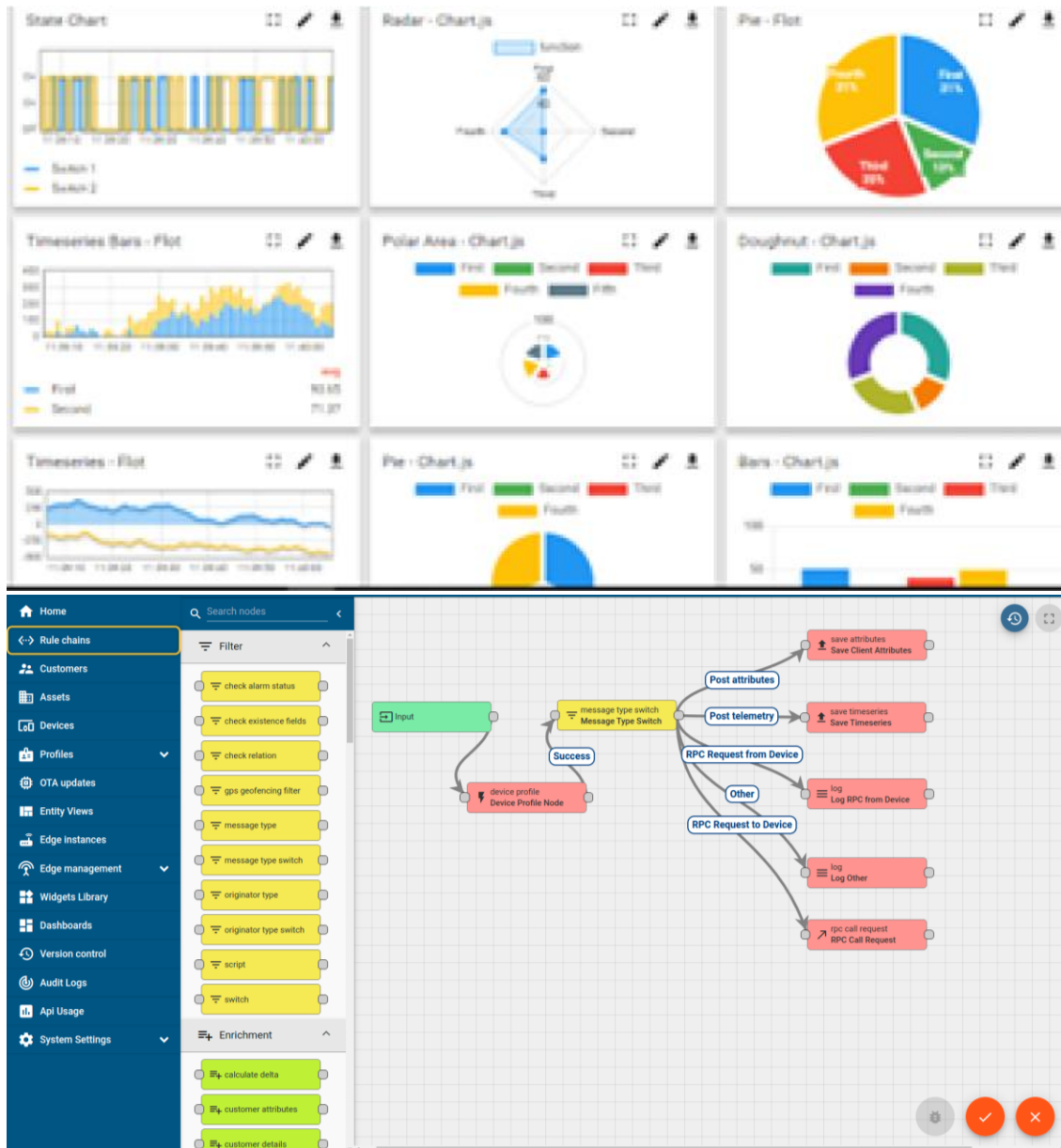
i. UCT IoT Platform ()

UCT Insight is an IOT platform designed for quick deployment of IOT applications on the same time providing valuable “insight” for your process/business. It has been built in Java for backend and ReactJS for Front end. It has support for MySQL and various NoSql Databases.

- It enables device connectivity via industry standard IoT protocols - MQTT, CoAP, HTTP, Modbus TCP, OPC UA
- It supports both cloud and on-premises deployments.

It has features to

- Build Your own dashboard
- Analytics and Reporting
- Alert and Notification
- Integration with third party application (Power BI, SAP, ERP)
- Rule Engine



FACTORY WATCH

ii. Smart Factory Platform ()

Factory watch is a platform for smart factory needs.

It provides Users/ Factory

- with a scalable solution for their Production and asset monitoring
- OEE and predictive maintenance solution scaling up to digital twin for your assets.
- to unleash the true potential of the data that their machines are generating and helps to identify the KPIs and also improve them.
- A modular architecture that allows users to choose the service that they want to start and then can scale to more complex solutions as per their demands.

Its unique SaaS model helps users to save time, cost and money.



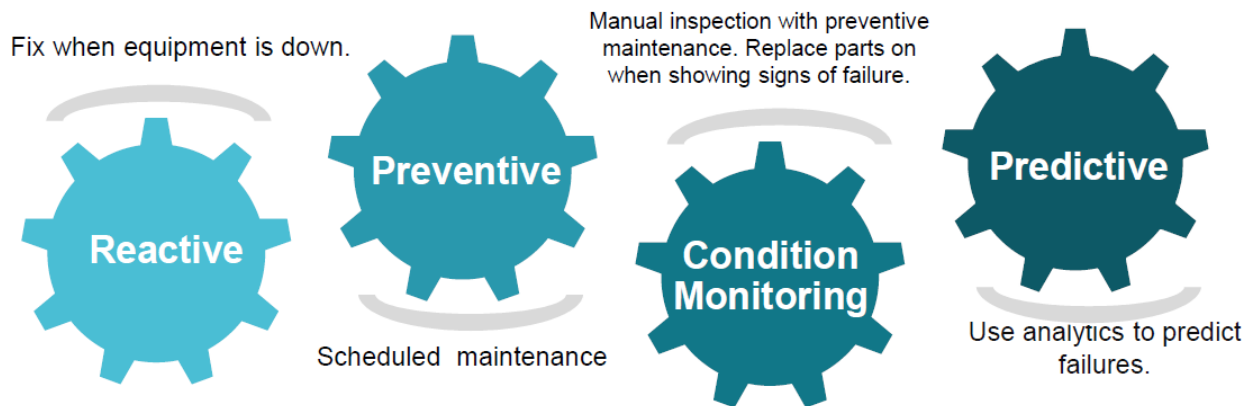


iii. LoRaWAN based Solution

UCT is one of the early adopters of LoRaWAN technology and providing solution in Agritech, Smart cities, Industrial Monitoring, Smart Street Light, Smart Water/ Gas/ Electricity metering solutions etc.

iv. Predictive Maintenance

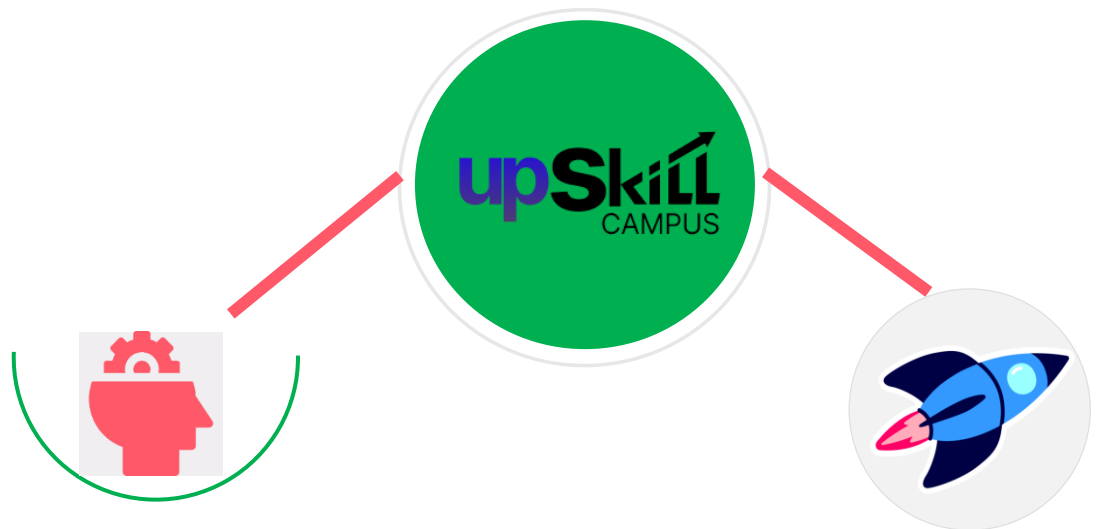
UCT is providing Industrial Machine health monitoring and Predictive maintenance solution leveraging Embedded system, Industrial IoT and Machine Learning Technologies by finding Remaining useful life time of various Machines used in production process.



2.2 About upskill Campus (USC)

upskill Campus along with The IoT Academy and in association with Uniconverge technologies has facilitated the smooth execution of the complete internship process.

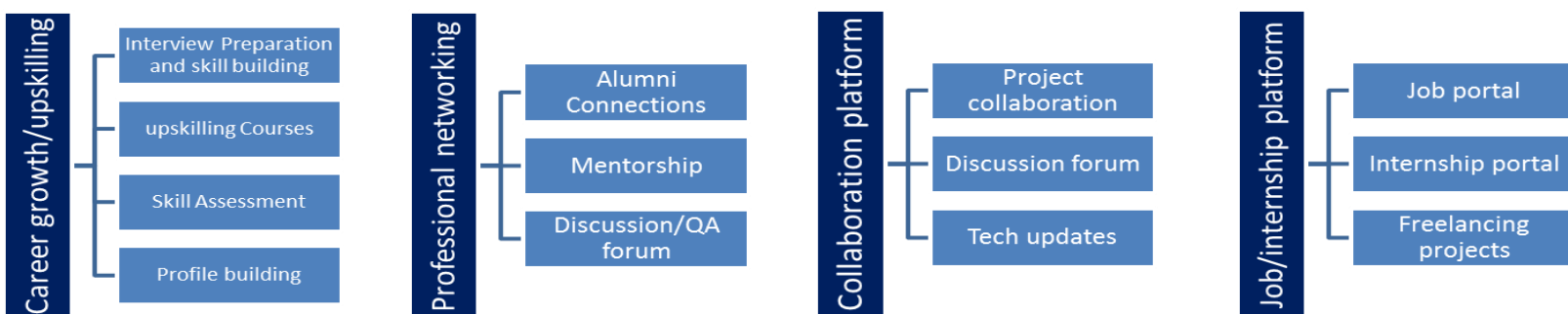
USC is a career development platform that delivers **personalized executive coaching** in a more affordable, scalable and measurable way.



Seeing need of upskilling in self paced manner along-with additional support services e.g. Internship, projects, interaction with Industry experts, Career growth Services

upSkill Campus aiming to upskill 1 million learners in next 5 year

<https://www.upskillcampus.com/>



2.3 The IoT Academy

The IoT academy is EdTech Division of UCT that is running long executive certification programs in collaboration with EICT Academy, IITK, IITR and IITG in multiple domains.

2.4 Objectives of this Internship program

The objective for this internship program was to

- get practical experience of working in the industry.
- to solve real world problems.
- to have improved job prospects.
- to have Improved understanding of our field and its applications.
- to have Personal growth like better communication and problem solving.

2.5 Reference

1. **upskill Campus**, *Embedded Systems and Internet of Things Internship Learning Materials*, 2026.
2. **UniConverge Technologies Pvt. Ltd. (UCT)**, *Industrial Internship Training Material*, 2026.
3. **Espressif Systems**, *ESP32 Series Datasheet*. Available:
<https://www.espressif.com/en/products/socs/esp32>

4. **Arduino. Arduino Documentation.** <https://docs.arduino.cc/>

5. **Wokwi, Online ESP32 Simulator.** Available: <https://wokwi.com/>

2.6 Glossary

Terms	Acronym
Internet of Things	IoT
ESP32 Microcontroller	ESP32
Microcontroller Unit	MCU
General Purpose Input/Output	GPIO
Temperature and Humidity Sensor	DHT22
Light Dependent Resistor	LDR
Organic Light Emitting Diode	OLED
Light Emitting Diode	LED
Integrated Development Environment	IDE
Wireless Fidelity	Wi-Fi

3 Problem Statement

Agriculture is one of the largest consumers of freshwater, and traditional irrigation methods often result in excessive water usage due to manual operation and a lack of real-time environmental monitoring. Farmers may irrigate crops without considering factors such as soil moisture, rainfall, temperature, humidity, or daylight conditions, leading to water wastage, increased energy consumption, and reduced irrigation efficiency.

The objective of this project is to develop an **IoT-Based Precision Irrigation Management System Using ESP32** that automates irrigation decisions based on real-time sensor data. The system continuously monitors soil moisture, temperature, humidity, rainfall, and ambient light conditions to determine whether irrigation is required. By supplying water only when necessary, the system promotes efficient water utilization, minimizes manual intervention, and supports sustainable agricultural practices.

The proposed solution demonstrates how embedded systems and IoT technologies can be integrated to create an intelligent irrigation system capable of improving resource management while maintaining a simple and cost-effective design suitable for small-scale agricultural applications.

4 Existing and Proposed Solution

Existing Solution

Traditional irrigation systems are primarily manual or timer-based. In manual irrigation, farmers inspect field conditions and operate irrigation pumps based on personal judgment. Timer-based systems irrigate crops at fixed intervals regardless of changing environmental conditions. These methods often fail to consider important parameters such as soil moisture, rainfall, temperature, humidity, and daylight availability.

Limitations of Existing Systems

- High water wastage due to unnecessary irrigation.
- Dependence on continuous human monitoring and intervention.
- Inability to respond to changing environmental conditions.
- Increased energy consumption because pumps may operate even when irrigation is not required.
- Limited efficiency in water resource management.

Proposed Solution

The proposed solution is an **IoT-Based Precision Irrigation Management System Using ESP32** that automates irrigation based on real-time environmental data. The system continuously monitors soil moisture, temperature, humidity, rainfall, and ambient light using appropriate sensors. Based on predefined conditions programmed into the ESP32, the system determines whether irrigation is required.

The irrigation pump is activated only when:

- The soil moisture level indicates dry soil.
- Rain is not detected.
- It is daytime.
- Humidity is below the specified threshold.
- Temperature is above the specified threshold.

The system also provides real-time status information through an OLED display, while an LED represents the pump status and a buzzer alerts the user whenever the irrigation system is activated. The project was designed and tested using the Wokwi simulation platform, demonstrating the practical application of Embedded Systems and IoT technologies for efficient irrigation management.

Advantages of the Proposed Solution

- Reduces water wastage through intelligent irrigation.
- Minimizes manual intervention.
- Improves irrigation efficiency using multiple environmental parameters.
- Supports sustainable agricultural practices.
- Provides real-time monitoring and system status.
- Cost-effective and suitable for small-scale agricultural applications.

4.1 Code submission (GitHub link):

https://github.com/ashelnoronha/upskillCampus/blob/main/IoT_Precision_Irrigation.ino

4.2 Report submission (GitHub link):

https://github.com/ashelnoronha/upskillCampus/blob/main/PrecisionIrrigationSystem_Ashel_USC_UCT.pdf

5 Proposed Design/ Model

The proposed model is designed to automate irrigation by continuously monitoring environmental conditions using multiple sensors connected to an ESP32 microcontroller. The system collects real-time data from the soil moisture sensor, DHT22 temperature and humidity sensor, rain sensor, and LDR to determine whether irrigation is required. Based on the programmed decision logic, the ESP32 controls the irrigation pump, updates the OLED display with system information, and provides an audible alert when irrigation begins. The following sections present the overall system architecture, detailed hardware design, operational flow, and system interfaces.

5.1 High Level Diagram

The high-level design illustrates the overall architecture of the proposed IoT-Based Precision Irrigation Management System. The ESP32 serves as the central controller and receives input from various sensors, including the soil moisture sensor, DHT22 sensor, rain sensor, and LDR. Based on the collected sensor data, the ESP32 processes the irrigation logic and controls the output devices such as the OLED display, pump indicator (LED), and buzzer. This architecture enables real-time monitoring and intelligent irrigation decisions.

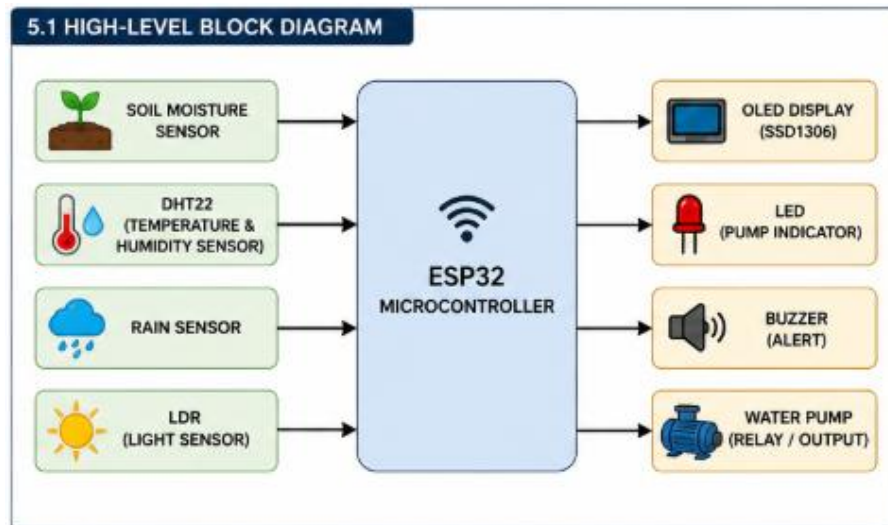


Figure 1: HIGH LEVEL DIAGRAM OF THE SYSTEM

5.2 Low Level Diagram

The low-level design represents the detailed hardware connections between the ESP32 and the various input and output devices used in the system. It shows the interfacing of sensors, display module, LED, and buzzer with their respective GPIO pins. This design provides a clear understanding of the hardware implementation of the proposed irrigation system.

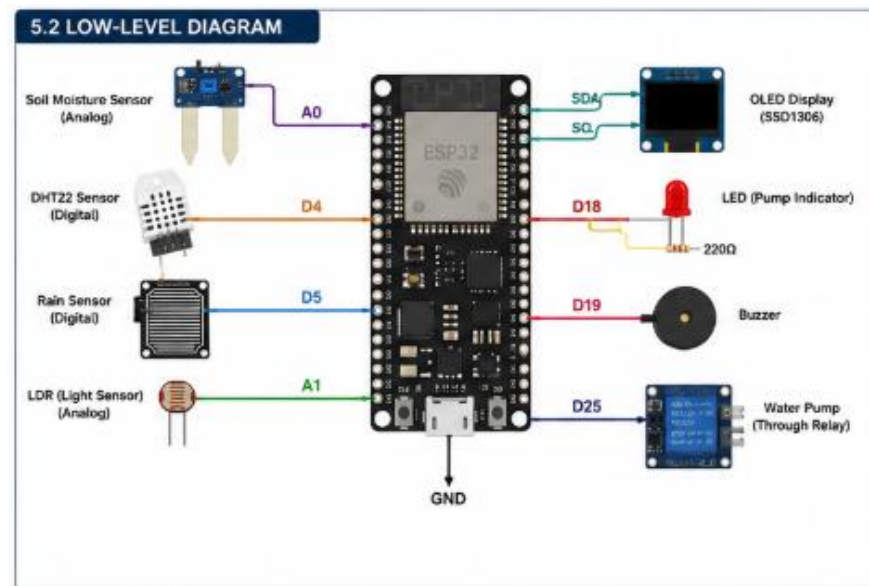


Figure 2: LOW LEVEL DIAGRAM OF THE SYSTEM

5.3 Interfaces

The flowchart illustrates the operational sequence of the proposed system. It shows how the ESP32 initializes the sensors, continuously monitors environmental parameters, evaluates the irrigation conditions, controls the pump accordingly, updates the OLED display, and repeats the process for continuous monitoring.

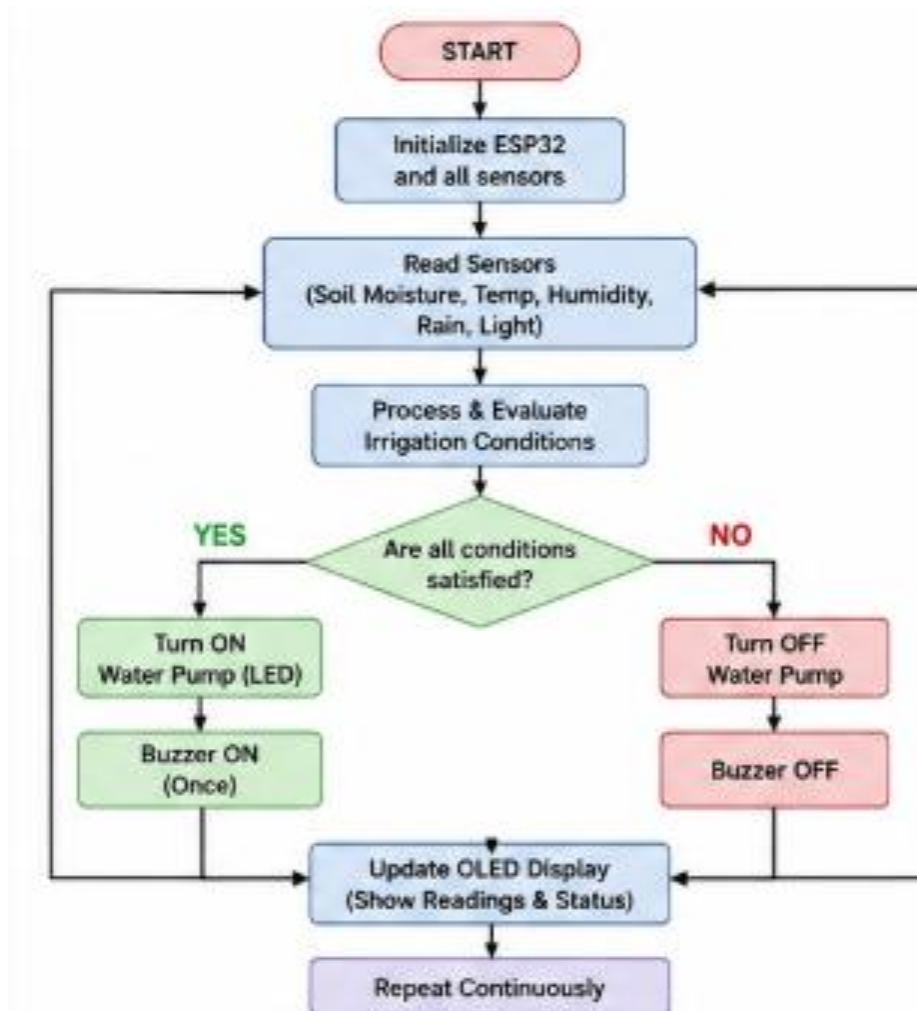


Figure 3: DESIGN FLOW OF THE SYSTEM

5.4 Simulation Model

The proposed IoT-Based Precision Irrigation Management System was designed and validated using the Wokwi online simulation platform. The simulation integrates the ESP32 microcontroller with sensors including the DHT22, soil moisture sensor, rain sensor, and LDR, along with output devices such as the OLED display, LED, and buzzer. The simulation environment enabled testing of the complete irrigation logic under different environmental conditions before real-world implementation.

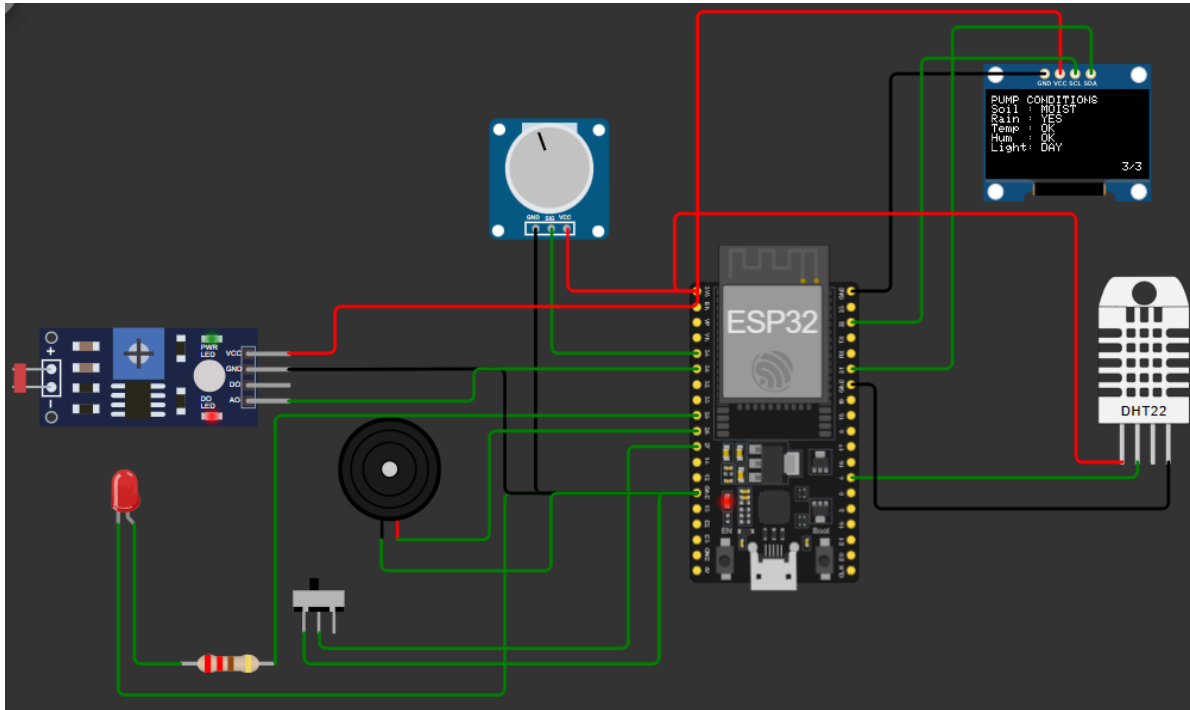


Figure 4: Wokwi Simulation of the IoT-Based Precision Irrigation Management System

6 Performance Test

The proposed IoT-Based Precision Irrigation Management System was tested using the Wokwi simulation platform to verify its functionality under different environmental conditions. Individual sensors were tested by varying their input values to ensure that the ESP32 correctly interpreted the data and executed the programmed irrigation logic.

The soil moisture sensor was adjusted to simulate dry, moist, and wet soil conditions. Temperature and humidity values were modified through the DHT22 sensor, while the rain sensor and LDR were used to simulate different weather and lighting conditions. The OLED display was monitored to verify that real-time sensor readings and system status were displayed correctly. The LED served as the irrigation pump indicator, and the buzzer was tested to ensure that it produced a single alert whenever the pump was activated.

The system performed as expected during simulation. Irrigation was initiated only when all predefined conditions were satisfied:

- Soil status was **Dry**.

- Rain was **not detected**.
- It was **daytime**.
- Temperature was **greater than 30°C**.
- Humidity was **less than 85%**.

Whenever any one of these conditions was not satisfied, the irrigation pump remained OFF. This confirmed that the decision-making logic implemented in the ESP32 functioned correctly and prevented unnecessary irrigation.

6.1 Test Cases

Test Case	Input Conditions	Expected Result	Observed Result
1	Dry soil, no rain, daytime, Temp > 30°C, Humidity < 85%	Pump ON	Passed
2	Wet soil	Pump OFF	Passed
3	Rain detected	Pump OFF	Passed
4	Nighttime	Pump OFF	Passed
5	Humidity $\geq$ 85%	Pump OFF	Passed
6	Temperature $\leq$ 30°C	Pump OFF	Passed

6.2 Test Procedure

The performance of the proposed IoT-Based Precision Irrigation Management System was evaluated using the Wokwi simulation platform. The ESP32 was programmed with the irrigation control logic, and all sensors and output devices were connected according to the designed circuit.

During testing, the sensor values were varied manually to simulate different environmental conditions. The soil moisture sensor was adjusted to represent dry, moist, and wet soil conditions. Temperature and humidity values were modified using the DHT22 sensor, while the rain sensor and LDR were toggled to simulate rainfall and day/night conditions, respectively.

For each test case, the OLED display was observed to verify that sensor readings and system status were updated correctly. The LED was monitored to confirm the pump status, and the buzzer was checked to ensure that it generated a single alert whenever irrigation was initiated.

The observed outputs were then compared with the expected outcomes to validate the correctness of the control logic.

6.3 Performance Outcome

- All sensors responded correctly during simulation.
- OLED displayed real-time sensor readings and irrigation status.
- Pump indicator (LED) activated only when all conditions were satisfied.
- Buzzer generated a one-time alert when irrigation started.
- The system successfully reduced unnecessary irrigation by considering multiple environmental parameters.

7 My Learnings

The internship provided me with valuable knowledge and practical experience in the fields of Embedded Systems and the Internet of Things (IoT). Throughout the training, I gained a strong understanding of IoT architecture, communication protocols, embedded hardware, sensors, actuators, and microcontroller programming. These concepts helped me understand how intelligent systems are designed to solve real-world problems.

During the development of the **IoT-Based Precision Irrigation Management System Using ESP32**, I learned how to interface multiple sensors with the ESP32 microcontroller, process real-time sensor data, and implement decision-making logic for automation. I also gained practical experience in programming using the Arduino IDE and testing embedded applications using the Wokwi simulation platform.

The internship enhanced my technical skills in circuit design, debugging, embedded software development, and system testing. It also improved my problem-solving abilities by requiring me to analyze different environmental conditions and implement efficient irrigation logic based on multiple sensor inputs.

Apart from technical knowledge, the internship strengthened my documentation, project planning, and analytical skills. It demonstrated the importance of integrating theoretical concepts with practical implementation to develop reliable and efficient IoT-based solutions.

Key Skills Acquired

- Embedded Systems Programming using ESP32
- Internet of Things (IoT) Fundamentals
- Sensor Interfacing and Data Acquisition
- Arduino IDE Programming
- Wokwi Simulation and Testing
- Embedded System Debugging
- Circuit Design and Hardware Integration
- IoT-Based Automation
- Technical Documentation and Report Writing
- Problem Solving and Analytical Thinking

8 Future Work Scope

The proposed IoT-Based Precision Irrigation Management System demonstrates an effective approach to automating irrigation using real-time environmental monitoring. Although the current system was developed and tested in a simulation environment, it can be further enhanced to support real-world agricultural applications.

Future improvements that can be incorporated into the system include:

- Implementing the system on physical hardware for deployment in agricultural fields.
- Integrating cloud platforms such as Firebase, ThingSpeak, or Blynk for remote monitoring and data storage.
- Developing a mobile or web application to allow farmers to monitor sensor data and control irrigation remotely.
- Incorporating weather forecasting APIs to prevent irrigation before expected rainfall.
- Using solar-powered energy sources to improve energy efficiency and enable deployment in remote locations.

- Adding GSM or Wi-Fi notification features to alert users about irrigation status or abnormal environmental conditions.
- Employing machine learning algorithms to analyze historical sensor data and optimize irrigation schedules based on crop requirements.
- Expanding the system to support multiple irrigation zones for large-scale farming applications.

The implementation of these enhancements would improve the intelligence, scalability, and efficiency of the irrigation system while promoting sustainable water management and smart agriculture.